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## Chapter 4

### Capacity Requirements Planning



# Capacity Requirements Planning

## Introduction

Capacity Requirements Planning (CRP) measures the resources required against the resources available to ensure that the correct balance of manufacturing capacity is available to produce a given production schedule.

The Capacity Requirements Planning system takes suggested orders from MPS and MRP and actual orders from Production Control, in any combination of suggested, planned, confirmed, released and/or active orders and schedules.

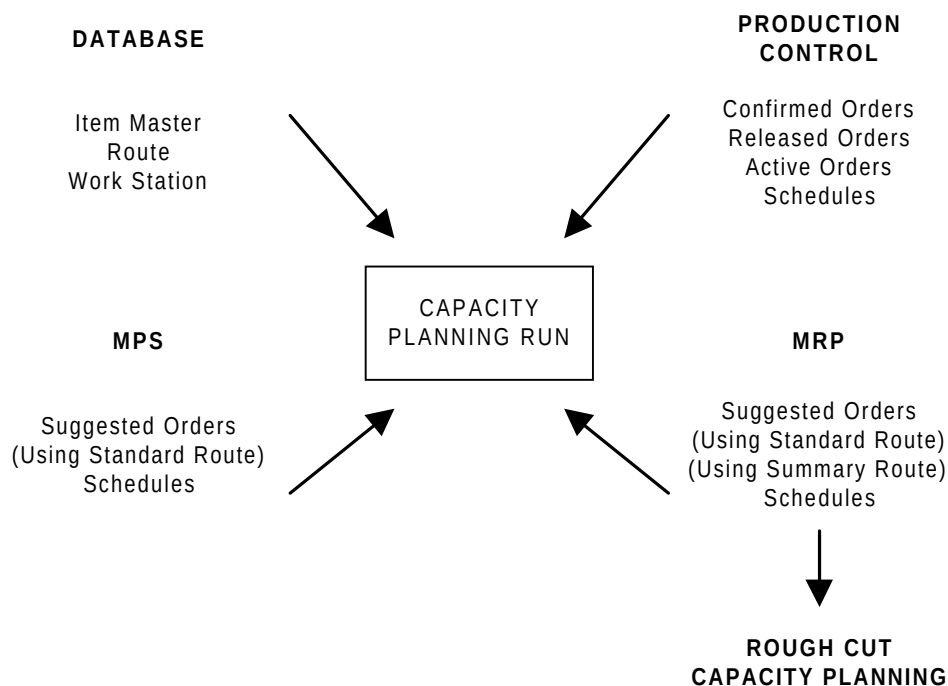
Using the start dates of the selected orders and taking into account any move days or queue time, the system calculates when each job is to be performed at each work centre and accumulates the load on each individual work centre.

The available capacity is calculated from the standard capacity for each work centre. In addition, allowance can be made for work centre down time caused by planned maintenance, unreliability, etc.

The output from capacity planning may be viewed on the screen in various ways or printed out.

The standard reporting unit for capacity planning is in weeks.

Data Input To The Capacity Planning Run



## Work Station Planned Down Time



### 1/CPM

You can enter the work station down time. This must be done before the capacity planning run, so that its effect can be taken into account in the run.



Select Work Station Planned Down Time on the menu to display the first Work Station Planned Down Time screen.

## Work Station Planned Down Time Screen 1



Select Work Station Planned Down Time on the menu to display the first Work Station Planned Down Time screen.

### Fields:

#### Model

Enter the CRP model within which down time is to be recorded.

#### Year

Enter the year to which the down time relates. A calendar for the year must have been created via Calendar Maintenance.

#### Work Station

Enter the work station for which the down time is to be recorded.



Select **F22=Include Actual Down Time**, to display the actual down time as well as the planned down time.



Select **F8=Update** on the first Work Station Planned Down Time screen to display the second screen.

## Work Station Planned Down Time Screen 2 (F22)



Select **F8=Update** on the first Work Station Planned Down Time screen to display the second screen.

| CP010                                                                                                |          | Z1 - GB Demonstration Company |                  | USER GBJBAMS | 8/12/95 |
|------------------------------------------------------------------------------------------------------|----------|-------------------------------|------------------|--------------|---------|
| Work Station Planned Down Time                                                                       |          |                               |                  |              |         |
| Work Station                                                                                         | WS01     | Work Station 01               |                  |              |         |
| Model                                                                                                | 01       |                               |                  |              |         |
| Year                                                                                                 | 95       |                               |                  |              |         |
| Week                                                                                                 | Start    | Planned Down Time             | Actual Down Time |              |         |
| 1/95                                                                                                 | 2/01/95  | 0:00                          | 0:00             |              |         |
| 2/95                                                                                                 | 9/01/95  | 0:00                          | 0:00             |              |         |
| 3/95                                                                                                 | 16/01/95 | 0:00                          | 0:00             |              |         |
| 4/95                                                                                                 | 23/01/95 | 0:00                          | 0:00             |              |         |
| 5/95                                                                                                 | 30/01/95 | 0:00                          | 0:00             |              |         |
| 6/95                                                                                                 | 6/02/95  | 0:00                          | 0:00             |              |         |
| 7/95                                                                                                 | 13/02/95 | 0:00                          | 0:00             |              |         |
| 8/95                                                                                                 | 20/02/95 | 0:00                          | 0:00             |              |         |
| 9/95                                                                                                 | 27/02/95 | 0:00                          | 0:00             |              |         |
| 10/95                                                                                                | 6/03/95  | 0:00                          | 0:00             |              |         |
| 11/95                                                                                                | 13/03/95 | 0:00                          | 0:00             |              |         |
| 12/95                                                                                                | 20/03/95 | 0:00                          | 0:00             |              |         |
| 13/95                                                                                                | 27/03/95 | 0:00                          | 0:00             | +            |         |
| F3=Exit   F8=Update   F11=Delete   F12=Select<br>09-42   SF   MM   KS   IM   II   KENB20A   PET166S2 |          |                               |                  |              |         |

### Fields:

#### Down Time

This screen allows the planned and actual down time to be entered in hours and minutes (HH:MM) for each week.



**Warning:** Do not attempt to enter the colon between the hours and minutes. Enter just four numbers, that is, for one hour and 25 minutes, enter 0125 and press Enter.



**Note:** If **Enter** was pressed on the last screen, instead of using **F22=Include Actual Down Time**, the same screen is displayed but without the actual down time being shown.

# Run CRP



## 11/CPM

Capacity Requirements Planning (CRP). The capacity planning run is a series of calculations undertaken by the system, using the data derived from the other Production applications. The objective is to determine if sufficient capacity is available to complete a given production plan. Rough cut capacity planning is identical planning, except that it is restricted to MPS items and may be a specially defined summary route. Factors used in the processing logic are:

### Starting Time

This is calculated from the due date of the order, offset by any move days or queue time, plus the duration time for each operation as defined in the route details and controlled by the selection method in the company profile. This determines the start time for each order.

### Weekly Load

Having calculated the start time and duration of each order, the duration is spread over the available working days. To do this, it is necessary to know how much time is available in each week. The available time is calculated as:

- Standard shift length x Number of shifts x Number of working days in the week

If the amount of time available in the week is greater than the calculated operation duration time, then the work is scheduled for that week.

If the amount of time available in the week is less than the calculated operation time then the job is scheduled to take place in the following week or weeks.



*Note: This process is carried out order by order.*

In carrying out the weekly scheduling, the system assumes infinite capacity is available.

Only on enquiries and reports does the system compare the total load with the total capacity.

### Available Capacity

This is calculated for each work centre, for each week as:

Standard Capacity x Number of Shifts x Working Days

This capacity is called the Budget Capacity on enquiries and reports.



*Note: Maximum capacity is also calculated, for reference purposes only. It is calculated in the same way as budget capacity but is called maximum capacity.*



Select Run CRP on the menu to display the Capacity Planning Run screen.

```

CP001                Z1 - GB Demonstration Company    USER GBJBAMS    8/12/95
                                                14:21:38

Capacity Planning Run

Model.....?.. █
Start Week..... 49/95
End Week..... 0/00
Base Plan On...Suggested Supply.. 0
                  Planned Supply.... 0
                  Confirmed Supply.. 0
                  Released/Active... 0

Last run details Status - Completed Normally
Requested By GBJBAMS      At QPADEV0010
Run Date 4/12/95      Time 17:36:29
Model 01
Start Week 49/95
End Week 25/96
Base Plan On...Suggested Supply 1
                  Planned Supply 1
                  Confirmed Supply 1
                  Released Supply 1

1=Yes
F3=Exit  F4=Prompt  F8=Submit Planning Run
05-53   SF      MK      KS      IM      II KENB20A PET166S2

```

## Fields:

### Model

Enter the CRP model to be generated. If the model is to utilise suggested production orders and/or schedules, the code entered must have an equivalent MPS and/or MRP model with the same code. Entering a CRP model without an equivalent MPS/MRP model does not extract any suggested orders or schedules. CRP, MPS and MRP models can be used to assess the effect different production plans have on resource capacity.

### Start Week

Enter the start date as a week number (WWYY). This week must exist in the calendar. This date is not normally earlier than the date on which the run is being carried out and prior periods cannot be enquired or reported on. It is advisable to leave this week as the current week.

### End Week

Enter the last date for which you wish to see work station loading. This figure represents the planning horizon. The earlier this date is, the less time the capacity planning run takes, and so it is sensible to make the planning period as short as is convenient for your particular planning requirements.

### Suggested Supply

Enter 1 to include production orders/schedules suggested by MPS and MRP in the capacity planning run. If you do this, the model code entered above must be valid in MPS and MRP.

### Planned Supply

Planned orders are created in Production Order Control, and do not have firm operational timings or material requirements. The system explodes planned orders through the designated planning route for the parent item to ascertain the operation loading.

This option does not include any schedule items.

## Confirmed Supply

Confirmed orders are created in Production Order Control and represent a firm commitment to make. They have firm operational timings which are extracted to calculate work station loads.

Confirmed schedules (similar actual commitment to make) are also included.

## Released/Active Supply

Enter 1 to include released and active orders in the run. These two types of supply are included, together as they are both of use in short-term planning.



*Note: The system places some restrictions on other production activities while the CRP run is taking place. For example, on selecting a model, you may be warned that is in progress. The status of the model code of the last run selection appears on the bottom half of the run selection screen, once at least one run selection has been made.*



*Note: It is advisable to make sure that all relevant calendars extend a little further than the end date you select. An error condition may occur if a date is encountered which does not exist on the calendar.*



*Note: For each individual order, finite scheduling is applied; for the total loading, infinite scheduling is applied; therefore loadings in excess of 100% may be produced.*



# Barchart



**21/CPM**



Select Barchart on the menu to display the  
Capacity Planning Enquiry Bar Chart screen.

## Capacity Planning Enquiry Bar Chart Screen



Select Barchart on the menu to display the Capacity Planning Enquiry Bar Chart screen.

|                                                         |                               |                                           |          |
|---------------------------------------------------------|-------------------------------|-------------------------------------------|----------|
| CP040                                                   | Z1 - GB Demonstration Company | USER GBJBAMS                              | 8/12/95  |
| Capacity Planning Enquiry - Bar Chart                   |                               |                                           | 14:47:24 |
| Model.....? <input type="text"/>                        |                               |                                           |          |
| Enquiry type..... 1                                     |                               | (1=Work Station)<br>(2=Work Centre)       |          |
| Work Station/ Work Centre.....? <input type="text"/>    |                               |                                           |          |
| Start week (current)..... 49/95                         |                               |                                           |          |
| End week..... 0/00                                      |                               |                                           |          |
| Review budget method..... 1                             |                               | (1=Standard)<br>(2=Standard x efficiency) |          |
| Include Supply Status                                   |                               |                                           |          |
| Suggested..... -                                        |                               | (1=Yes)                                   |          |
| Planned..... -                                          |                               | (1=Yes)                                   |          |
| Confirmed..... -                                        |                               | (1=Yes)                                   |          |
| Released/active... -                                    |                               | (1=Yes)                                   |          |
| F3=Exit F4=Prompt F10=Summary Enquiry                   |                               |                                           |          |
| 06-48 <input type="text"/> MW KS IM II KENB20A PET166S2 |                               |                                           |          |

### Fields:

#### Model

Enter the required model code. This must be the model for which the last CRP run was successfully run.

#### Enquiry Type

You can enter the following options:

- 1 - Enquire by work station
- 2 - Enquire by work centre.

#### Work Station/Work Centre

Enter the work station or work centre on which you wish to enquire, depending on the entry made above.

#### Start Week

The start week for the enquiry must be the current week. So the current week is displayed here.

#### End Week

Enter the week number for the last week which you require for inclusion in the enquiry.

#### Review Budget

Comparisons between scheduled loads and capacity at work stations can be made using one of two methods; the result of which is referred to and subsequently shown as budgeted capacity. You can enter the following options:

- 1 - To compare to standard capacity. The system assumes 100% efficiency; that is, all the capacity is used.
- 2 - To compare with standard effective capacity; that is, with the standard work station efficiency factor applied. (Standard capacity \* standard efficiency factor).

#### **Include Supply Status**

Enter 1 against each status of supply that you wish to include. Any combination of status' included in the last CRP run may be selected.



Select **F18=Summary Enquiry** on the Capacity Planning Enquiry Bar Chart screen to display the Capacity Planning Enquiry screen.

#### **Functions:**

##### **F18**

Summary Enquiry. Allows you to toggle between the summary and barchart enquiries screens.

## Capacity Planning Enquiry Screen (F18)



Select **F18=Summary Enquiry** on the Capacity Planning Enquiry Bar Chart screen to display the Capacity Planning Enquiry screen.

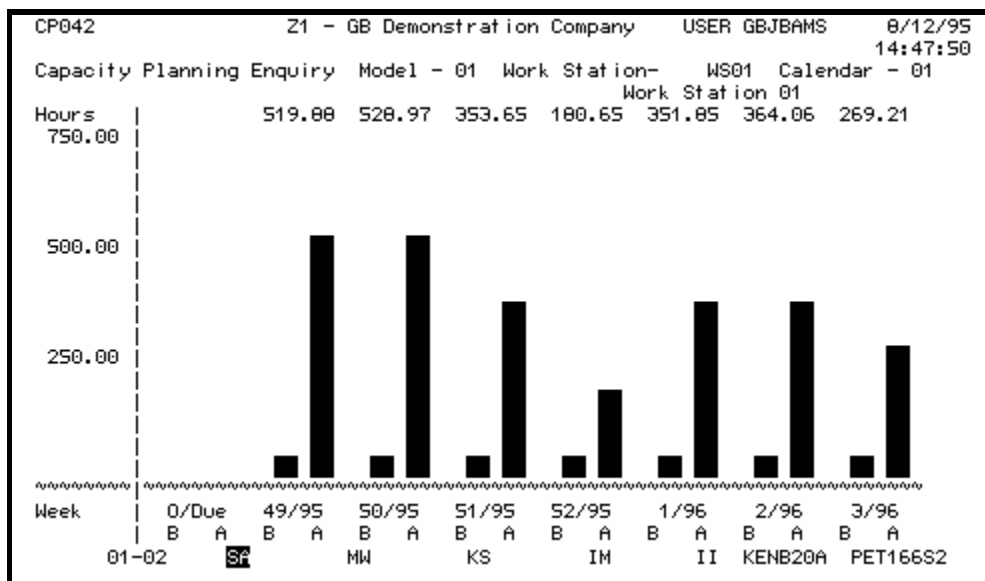
The vertical scale of the barchart is in hours. The horizontal scale shows the week number.

The first column shown any overdue work, followed by the current week and then subsequent weeks. For each week, there are two columns: the left hand column shows the Budgeted capacity and the right hand column shows the Actual number of hours scheduled, for the work station or work centre for that week.

The overdue column shows the amount of work outstanding.

The numbers above the columns show the number of hours work scheduled for each week.

Later and earlier weeks can be displayed by using the Page Up and Page Down keys.



# Summary



**22/CPM**



Select Summary on the menu to display the Capacity Planning Enquiry Summary screen.

## Capacity Planning Enquiry Summary Screen



Select Summary on the menu to display the Capacity Planning Enquiry Summary screen.

|                                        |                               |                                           |          |
|----------------------------------------|-------------------------------|-------------------------------------------|----------|
| CP040                                  | Z1 - GB Demonstration Company | USER GBJBAMS                              | 8/12/95  |
| Capacity Planning Enquiry - Summary    |                               |                                           | 14:22:10 |
| Model.....?.. _                        |                               |                                           |          |
| Enquiry type..... 1                    |                               | (1=Work Station)<br>(2=Work Centre)       |          |
| Work Station/ Work Centre.....?.. _    |                               |                                           |          |
| Start week (current)..... 49/95        |                               |                                           |          |
| End week..... 0/00                     |                               |                                           |          |
| Review budget method..... 1            |                               | (1=Standard)<br>(2=Standard x efficiency) |          |
| Include Supply Status                  |                               |                                           |          |
| Suggested..... _                       |                               | (1=Yes)                                   |          |
| Planned..... _                         |                               | (1=Yes)                                   |          |
| Confirmed..... _                       |                               | (1=Yes)                                   |          |
| Released/active... _                   |                               | (1=Yes)                                   |          |
| F3=Exit F4=Prompt F10=Barchart Enquiry |                               |                                           |          |
| 06-48 SF MM KS IM II KENB20A PET166S2  |                               |                                           |          |

### Fields:

#### Model

Enter the required model code. This must be model for which the last CRP run was successful.

#### Enquiry Type

You can enter the following options:

- 1 - Enquire by work station
- 2 - Enquire by work centre.

#### Work Station/Work Centre

Enter the work station or work centre on which you wish to enquire, depending on the entry made above.

#### Start Week

The start week for the enquiry must be the current week. So the current week is displayed here.

#### End Week

Enter the week number for the last week which you require for inclusion in the enquiry.

#### Review Budget

Comparisons between scheduled loads and capacity at work stations can be made using one of two methods; the result of which is shown as budgeted capacity. You can enter the following options:

- 1 - To compare to standard capacity. The system assumes 100% efficiency; that is, all the capacity is used.
- 2 - To compare with standard effective capacity; that is, with the standard work station efficiency factor applied. (Standard capacity x standard efficiency factor).

#### **Include Supply Status**

Enter 1 against each status of supply that you wish to include. Any combination of status' included in the last CRP run may be selected.



Press **Enter** on the Capacity Planning Enquiry Summary screen to display the first Capacity Planning Enquiry screen.

#### **Functions:**

##### **F18**

Barchart Enquiry. Allows you to toggle between the summary and barchart enquiries.

## Capacity Planning Enquiry Screen 1

The top section of this screen displays the work station or work centre selected. Where an individual work station has been selected, details for that station are displayed. Where a work centre has been selected, a list of the work stations in that group is displayed.

Each column represents the available capacity and the requirements for a particular week. The relevant week number is displayed above each column.

The first row gives the Maximum capacity of the work station (or centre). This is the total of the maximum shift capacities, held on the work station record, multiplied by the number of working days available in each week.

The second row gives the Budget capacity for each week. These figures represent the expected available capacity. This is the total of the standard shift capacities multiplied by the number of days available each week. This figure is then adjusted for any planned down time, entered for the relevant work station(s). If budget method 2 was selected on the previous screen, this is then multiplied by the standard efficiency percentage from the work station record.

The third row gives the Planned work load, the total number of hours that the CRP run has calculated to be scheduled for each week. This includes all operations that fall in that week according to the start dates on the orders regardless of any constraints due to capacity limitations.

The fourth row gives the week Loading percentage. This is calculated separately for each week in the enquiry.

The last row gives the Average loading or cumulative loading percentage. The total available capacity and the capacity requirements are totalled from the first week and an average loading value is given. This enables you to ascertain whether the overall loading is reasonable. The Page Up and Page Down keys can be used to display earlier and later weeks.



Press **Enter** on the Capacity Planning Enquiry Summary screen to display the first Capacity Planning Enquiry screen.



|                                      |                               |         |            |            |                  |                   |          |
|--------------------------------------|-------------------------------|---------|------------|------------|------------------|-------------------|----------|
| CP041                                | Z1 - GB Demonstration Company | USER    | GBJBAMS    | 8/12/95    |                  |                   |          |
|                                      |                               |         |            | 14:44:13   |                  |                   |          |
| Capacity Planning Enquiry Model - 01 |                               |         |            |            |                  |                   |          |
| Work Station                         | Queue time                    | Shift   | Max Length | Std Length | Maximum Capacity | Standard Capacity |          |
| WS01                                 |                               | 1       | 8.0        | 8.0        | 8.0000           | 8.0000            |          |
| Work Station 01                      |                               | 2       |            |            |                  |                   |          |
| Calendar 01                          | Std.Eff.                      | 3       |            |            |                  |                   |          |
| 0/Due                                |                               | 49/95   | 50/95      | 51/95      | 52/95            | 1/96              | 2/96     |
| Max. capacity...                     | 40.00                         | 40.00   | 40.00      | 24.00      | 40.00            | 40.00             |          |
| Budget capacity.                     | 40.00                         | 40.00   | 40.00      | 24.00      | 40.00            | 40.00             |          |
| Planned load....                     | 519.88                        | 528.97  | 353.65     | 180.65     | 351.85           | 364.06            |          |
| Loading %.....                       | 1299.70                       | 1322.43 | 884.13     | 752.71     | 879.63           | 910.15            |          |
| Average Loading.                     | 1299.70                       | 1311.06 | 1168.75    | 1099.41    | 1051.63          | 1026.37           |          |
| 3/96                                 |                               |         |            |            |                  |                   |          |
| Max. capacity...                     | 40.00                         |         |            |            |                  |                   |          |
| Budget capacity.                     | 40.00                         |         |            |            |                  |                   |          |
| Planned load....                     | 269.21                        |         |            |            |                  |                   |          |
| Loading %.....                       | 673.03                        |         |            |            |                  |                   |          |
| Average Loading.                     | 972.83                        |         |            |            |                  |                   |          |
| F3=Exit F12=Previous F17=Detail      |                               |         |            |            |                  |                   |          |
| 01-02                                | SF                            | MM      | KS         | IM         | II               | KENB20A           | PET166S2 |



Select **F17=Detail** on the first Capacity Planning Enquiry screen to display the second screen.

## Functions:

### F17

Detail. Allows you to toggle between the summary and loading details.

## Capacity Planning Enquiry Screen 2 (F17)

This screen displays the loading details.



Select **F17=Detail** on the first Capacity Planning Enquiry screen to display the second screen.

|                                                  |            |                               |            |            |                  |                   |                     |             |
|--------------------------------------------------|------------|-------------------------------|------------|------------|------------------|-------------------|---------------------|-------------|
| CP041                                            |            | Z1 - GB Demonstration Company |            |            | USER GBJBAMS     |                   | 8/12/95<br>14:44:13 |             |
| Capacity Planning Enquiry Model - 01             |            |                               |            |            |                  |                   |                     |             |
| Work Station                                     | Queue time | Shift                         | Max Length | Std Length | Maximum Capacity | Standard Capacity |                     |             |
| WS01                                             |            | 1                             | 8.0        | 8.0        | 8.0000           | 8.0000            |                     |             |
| Work Station 01                                  |            | 2                             |            |            |                  |                   |                     |             |
| Calendar 01 Std.Eff.                             |            | 3                             |            |            |                  |                   |                     |             |
| Operation Details                                |            |                               | Work       |            |                  |                   |                     |             |
| Week                                             | Str.Date   | Supply                        | Oper       | Dur(Hrs)   | S                | Qty               | O/S                 | UoM Item    |
| 49/95                                            | 4/12/95    | SUGGEST                       | 10         | 36.19      | 0                |                   | 86.666              | EA MS003    |
| 49/95                                            | 4/12/95    | SUGGEST                       | 10         | 31.25      | 0                |                   | 125.000             | EA MPSCOPAR |
| 49/95                                            | 4/12/95    | SUGGEST                       | 10         | 87.16      | 0                |                   | 209.000             | EA MS001    |
| 49/95                                            | 4/12/95    | SUGGEST                       | 10         | 55.91      | 0                |                   | 134.000             | EA MS001    |
| 49/95                                            | 4/12/95    | Frm Sch                       | 10         | 88.00      | 0                |                   | 211.000             | EA MS002    |
| 49/95                                            | 4/12/95    | Frm Sch                       | 10         | 74.25      | 0                |                   | 178.000             | EA MS002    |
| 49/95                                            | 4/12/95    | SUGGEST                       | 10         | 78.15      | 0                |                   | 234.000             | EA MS004    |
| 49/95                                            | 4/12/95    | W000002                       | 10         | 8.33       | 2                |                   | 100.000             | EA MP0318   |
| 49/95                                            | 4/12/95    | W000011                       | 10         | 5.03       | 2                |                   | 100.000             | EA MR0309   |
| 49/95                                            | 4/12/95    | W000012                       | 10         | 7.53       | 2                |                   | 150.000             | EA MR0309   |
| 49/95                                            | 4/12/95    | W000013                       | 10         | 7.53       | 2                |                   | 150.000             | EA MR0309   |
| 49/95                                            | 4/12/95    | W000014                       | 10         | 100.00     | 2                |                   | 100.000             | EA MPSPAR   |
|                                                  |            |                               |            |            |                  |                   |                     |             |
| F3=Exit F12=Previous F17=Summary F22=Week detail |            |                               |            |            |                  |                   |                     |             |
| 01-02                                            |            | SE                            | MW         | KS         | IM               | II                | KENB20A             | PET166S2    |



Select **F22=Week Detail** on the second Capacity Planning Enquiry screen to display the third screen.

### Functions:

#### F17

Summary. Allows you to toggle between the summary and loading details.

#### F22

Week Detail. When you select this function, the fourth Capacity Planning Enquiry screen is displayed.

## Capacity Planning Enquiry Screen 3 (F22)



Select **F22=Week Detail** on the second Capacity Planning Enquiry screen to display the third screen.

You can display a second line of detail.

If a work station has been selected, the work station details are displayed at the top of the screen. If a work centre has been selected, the work centre and the work stations in the centre are displayed.

|                                                  |                               |         |            |            |                  |                   |          |         |          |      |
|--------------------------------------------------|-------------------------------|---------|------------|------------|------------------|-------------------|----------|---------|----------|------|
| CP041                                            | Z1 - GB Demonstration Company |         |            |            | USER GBJBAMS     | 8/12/95           |          |         |          |      |
|                                                  |                               |         |            |            |                  | 14:44:13          |          |         |          |      |
| Capacity Planning Enquiry Model - 01             |                               |         |            |            |                  |                   |          |         |          |      |
| Work Station                                     | Queue time                    | Shift   | Max Length | Std Length | Maximum Capacity | Standard Capacity |          |         |          |      |
| WS01                                             |                               | 1       | 8.0        | 8.0        | 8.0000           | 8.0000            |          |         |          |      |
| Work Station 01                                  |                               | 2       |            |            |                  |                   |          |         |          |      |
| Calendar 01                                      | Std.Eff.                      | 3       |            |            |                  |                   |          |         |          |      |
| Operation Details                                |                               |         | Work       |            |                  |                   |          |         |          |      |
| Week                                             | Str.Date                      | Supply  | Oper       | Dur(Hrs)   | S                | Ctr               | Quantity | O/S     | UoM      | Item |
| 49/95                                            | 4/12/95                       | SUGGEST | 10         | 36.19      | 0                |                   | 86.666   | EA      | MS003    |      |
| This week - Op.duration:                         |                               |         |            | 36.11      |                  | Qty:              | 86.666   |         | Set Up:  | .00  |
| 49/95                                            | 4/12/95                       | SUGGEST | 10         | 31.25      | 0                |                   | 125.000  | EA      | MPSCOPAR |      |
| This week - Op.duration:                         |                               |         |            | 31.25      |                  | Qty:              | 125.000  |         | Set Up:  |      |
| 49/95                                            | 4/12/95                       | SUGGEST | 10         | 87.16      | 0                |                   | 209.000  | EA      | MS001    |      |
| This week - Op.duration:                         |                               |         |            | 39.91      |                  | Qty:              | 95.784   |         | Set Up:  | .00  |
| 49/95                                            | 4/12/95                       | SUGGEST | 10         | 55.91      | 0                |                   | 134.000  | EA      | MS001    |      |
| This week - Op.duration:                         |                               |         |            | 39.91      |                  | Qty:              | 95.784   |         | Set Up:  | .00  |
| 49/95                                            | 4/12/95                       | Frm Sch | 10         | 88.00      | 0                |                   | 211.000  | EA      | MS002    |      |
| This week - Op.duration:                         |                               |         |            | 39.92      |                  | Qty:              | 95.808   |         | Set Up:  | .00  |
| 49/95                                            | 4/12/95                       | Frm Sch | 10         | 74.25      | 0                |                   | 178.000  | EA      | MS002    |      |
| This week - Op.duration:                         |                               |         |            | 39.92      |                  | Qty:              | 95.808   |         | Set Up:  | .0 + |
| F3=Exit F12=Previous F17=Summary F22=Week detail |                               |         |            |            |                  |                   |          |         |          |      |
| 01-02                                            |                               | SE      | MW         | KS         |                  | IM                | II       | KENB20A | PET166S2 |      |

### Fields:

#### Week

The number of the week in which this load falls.

#### Start Date

The date on which this operation is scheduled to begin.

#### Supply

Descriptive status of the supply from which this load arises. If a firm production order, this is the order number.

#### Operation

The operation sequence number of the operation which gives rise to this load.

#### Duration

The number of hours outstanding for this operation, at the beginning of the week.

#### S

Status. The current status of the supply:

0 for suggested order or schedule

1 for planned order

2 for firm order or schedule

3 for released order

4 for active order

### **Work Centre**

The work station to which this operation and load is related.

### **Quantity O/S**

The quantity outstanding for this operation, at the beginning of the week. This is calculated as the operation scheduled quantity less the operation quantity completed.

### **UOM**

The unit of measure in which the quantity outstanding is expressed.

### **Item**

The finished item.

### **This Week - Operation Duration**

The duration of the work scheduled to take place during this week, for this operation.

### **Quantity:**

The quantity scheduled to be completed this week, for this operation. This is calculated from the capacity available in the work centre during the week.

### **Set Up:**

The duration of any set-up scheduled for the operation for this week. This is calculated from the capacity available in the work station during the week.



Select **F3=Exit** to return to the menu.

### **Functions:**

#### **F17**

Summary. Allows you to toggle between the summary and loading details.

#### **F22**

Week Detail.

# Work Station Down Time Plan

Reports provide a printed copy of the results of any Capacity Planning run. Various selections are possible to allow the reports to be tailored to suit individual requirements.



## 31/CPM



Select Work Station Down Time Plan on the menu to display the Work Station Down Time Plan Report screen.

|                                                                                                                                                                                               |                               |              |          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------|----------|
| CP030                                                                                                                                                                                         | Z1 - GB Demonstration Company | USER GBJBAMS | 8/12/95  |
| Work Station Down Time Plan Report                                                                                                                                                            |                               |              | 14:48:20 |
| <p>Model code range from...?. ____ (Blank for first)</p> <p>to....?. ____ (Blank for last)</p> <p>Work Station range from.?. ____ (Blank for first)</p> <p>to....?. ____ (Blank for last)</p> |                               |              |          |
| <p>F3=Exit F4=Prompt</p> <p>07-38  MW KS IM II KENB20A PET166S2</p>                                        |                               |              |          |

## Fields:

### Model Range

Enter the range of model codes to be included.

### Work Station Range

Enter the range of work stations required for inclusion in the report. These appear in alphanumeric sequence on the report. Leave both fields blank to report on all work stations or enter the same code in both fields to report on a single work station.

# Work Station Capacity Requirements



## 32/CPM



Select Work Station Capacity Requirements on the menu to display the Capacity Report By Work Station screen.

|                                                                              |                               |              |          |
|------------------------------------------------------------------------------|-------------------------------|--------------|----------|
| CP033                                                                        | Z1 - GB Demonstration Company | USER GBJBAMS | 8/12/95  |
| Capacity Report - By Work Station                                            |                               |              | 14:48:45 |
| Model.....?.. <b>1</b>                                                       |                               |              |          |
| Review budget method..... <b>1</b> (1=Standard)<br>(2=Standard x efficiency) |                               |              |          |
| Start week (current)..... <b>49/95</b>                                       |                               |              |          |
| End week..... <b>0/00</b>                                                    |                               |              |          |
| From Work Station.....?.. <b>_____</b> (Blank for first)                     |                               |              |          |
| To Work Station.....?.. <b>_____</b> (Blank for last)                        |                               |              |          |
| Base plan on : Suggested supply.. <b>0</b> (1=Yes)                           |                               |              |          |
| Planned supply.... <b>0</b> (1=Yes)                                          |                               |              |          |
| Confirmed supply.. <b>0</b> (1=Yes)                                          |                               |              |          |
| Released/active... <b>0</b> (1=Yes)                                          |                               |              |          |
| Print operation details..... <b>1</b> (1=Yes)                                |                               |              |          |
| F3=Exit F4=Prompt                                                            |                               |              |          |
| 06-48 <b>SF</b> MW KS IM II KENB20A PET166S2                                 |                               |              |          |

## Fields:

### Model

Enter the model code.

### Review Budget Method

Comparisons between scheduled loads and capacity at work stations can be made using one of two methods; the result of which is shown as budgeted capacity. You can enter the following options:

1 - To compare to standard capacity. The system assumes 100% efficiency; that is all the capacity is used.

2 - To compare with the standard effective capacity; that is, with the standard work station efficiency factor applied. Calculated in the following way: Standard Capacity \* Standard Efficiency Factor.

### Start Week

The start week for the report must be the current week and is displayed as such on the screen.

### End Week

Enter the week number of the last week which you wish to be included in the report.

### From/To Work Station

Enter the range of work stations required for inclusion in the report.

**Base Plan On**

Enter 1 against each supply status that you wish to include in the report. It is possible to select any combination of status' which were included in the last Capacity Planning run for the selected model.

**Print Operation Details**

Enter 1, if you require operation details to be included below the summary table, in the report or leave blank to print a summary table only.

# Work Centre Capacity Requirements



## 33/CPM



Select Work Centre Capacity Requirements on the menu to display the Capacity Report By Work Centre screen.

The selection criteria on this screen are the same as for the Work Station Capacity Requirements report except a range is required for work centres instead of work stations.

|                                                                              |                               |              |          |
|------------------------------------------------------------------------------|-------------------------------|--------------|----------|
| CP033                                                                        | Z1 - GB Demonstration Company | USER GBJBAMS | 8/12/95  |
| Capacity Report - By Work Centre                                             |                               |              | 14:49:11 |
| Model.....?.. <b>1</b>                                                       |                               |              |          |
| Review budget method..... <b>1</b> (1=Standard)<br>(2=Standard x efficiency) |                               |              |          |
| Start week (current)..... <b>49/95</b>                                       |                               |              |          |
| End week..... <b>0/00</b>                                                    |                               |              |          |
| From Work Centre.....?.. _____ (Blank for first)                             |                               |              |          |
| To Work Centre.....?.. _____ (Blank for last)                                |                               |              |          |
| Base plan on : Suggested supply.. <b>0</b> (1=Yes)                           |                               |              |          |
| Planned supply.... <b>0</b> (1=Yes)                                          |                               |              |          |
| Confirmed supply.. <b>0</b> (1=Yes)                                          |                               |              |          |
| Released/active... <b>0</b> (1=Yes)                                          |                               |              |          |
| Print operation details..... <b>1</b> (1=Yes)                                |                               |              |          |
| F3=Exit F4=Prompt                                                            |                               |              |          |
| 06-48 <b>SP</b> MW KS IM II KENB20A PET166S2                                 |                               |              |          |

## Fields:

### Model

Enter the model code.

### Review Budget Method

Comparisons between scheduled loads and capacity at work centres can be made using one of two methods; the result of which is shown as budgeted capacity. You can enter the following options:

1 - To compare to standard capacity. The system assumes 100% efficiency; that is all the capacity is used.

2 - To compare with standard effective capacity; that is, with the standard work centre efficiency factor applied. (Standard capacity \* standard efficiency factor).

### Start Week

The start week for the report must be the current week and is displayed as such on the screen.

### End Week

Enter the week number of the last week which you wish to be included in the report.



**From/To Work Centre**

Enter the range of work centres to include in the report.

**Base Plan On**

Enter 1 by each supply status you want to include in the report. You can select any combination of status' which were included in the last Capacity Planning run for the selected model.

**Print Operation Details**

Enter 1, if you require operation details to be included below the summary table, in the report or leave blank to print a summary table only.